

Midterm Exam - Sample

Database 1 - Spring 2026

Name: _____ Total: 115 pts

Exam is out of 100 points. Maximum possible is 115 - any points above 100 count as bonus.

Dataset: GymPro Management System

GymPro tracks gym operations across four tables:

- **members** (registered members and their plans),
- **trainers** (instructors and their details),
- **classes** (scheduled fitness sessions),
- **checkins** (member attendance logs).

Schema

members

Column	Type	Key
member_id	INTEGER	PK
name	TEXT	
age	INTEGER	
gender	TEXT	
email	TEXT	
phone	TEXT	
join_date	DATE	
membership_type	TEXT	
monthly_fee	REAL	
status	TEXT	

trainers

Column	Type	Key
trainer_id	INTEGER	PK
name	TEXT	
age	INTEGER	
specialty	TEXT	
experience_years	INTEGER	
hourly_rate	REAL	
available_days	TEXT	
rating	REAL	

classes

Column	Type	Key
class_id	INTEGER	PK
class_name	TEXT	
trainer_id	INTEGER	FK → trainers.trainer_id
schedule_date	DATE	
start_time	TEXT	
duration_mins	INTEGER	
capacity	INTEGER	
room	TEXT	
difficulty_level	TEXT	

checkins

Column	Type	Key
checkin_id	INTEGER	PK
member_id	INTEGER	FK → members.member_id
class_id	INTEGER	FK → classes.class_id
checkin_date	DATE	
checkin_time	TEXT	
early_cancel	INTEGER	

Questions

Q1 (15 pts) - List all active members who joined after January 1st, 2024.

Return: name, membership_type, monthly_fee. Order by monthly_fee descending.

Q2 (15 pts) - Show each class name along with the trainer who teaches it.

Return: class_name, name (trainer), specialty (trainer).

Q3 (20 pts) - For each membership type, find the average monthly fee and total number of members. Only show groups with more than 5 members.

Return: membership_type, avg_fee, total_members

Q4 (20 pts) - Find all members who have never checked into any class.

Return: name

Q5 (20 pts) - Show each member's name, checkin date, and the number of checkins they have made up to and including that date. Only include members who have at least one checkin.

Return: name, checkin_date, running_total

Q6 (25 pts) - For each trainer, show total classes taught per month in 2024 and categorize their monthly workload as 'High' (8 or more), 'Medium' (4 to 7), or 'Low' (fewer than 4).

Return: name (trainer), month, total_classes, workload. Order by trainer name then month.